

R_{LEAD3} is not part of the measurement, because it is not in the input measurement path or in the reference input path.

1.4.2 IDAC Current Chopping

As described in the previous section, the two current sources must be matched to cancel the lead resistances of the RTD wires. Any mismatch in the two current sources may be minimized by using the multiplexer (MUX) to swap or *chop* the two current sources between the two inputs. Taking two measurements in each configuration and averaging the results reduces the effects of mismatched current sources.

Using the configuration from Figure 1-5, Equation 6 results in the first measurement. Figure 1-6 swaps IDAC1 and IDAC2, and Equation 9 results in the second measurement.

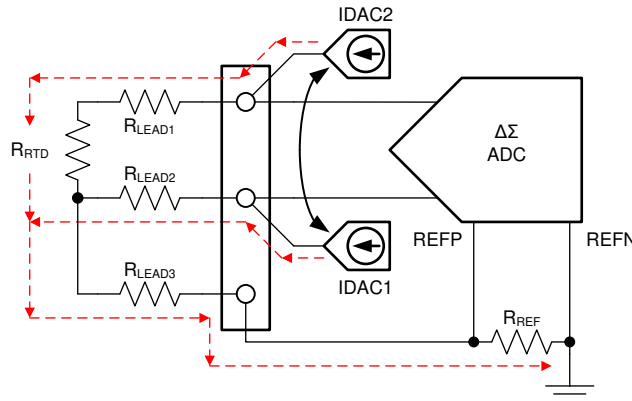


Figure 1-6. Swapping IDAC1 and IDAC2 to Chop the Measurement

$$\text{Output code} = 2^{23} \cdot [I_{IDAC2} \cdot (R_{LEAD1} + R_{RTD}) - (I_{IDAC1} \cdot R_{LEAD2})] / [(I_{IDAC1} + I_{IDAC2}) \cdot R_{REF}] \quad (9)$$

To chop the RTD measurement, we average the first and second measurements. Take Equation 6, add it to Equation 9 and then divide by two to average the result. This is shown in the following:

$$\text{Averaged output code} = \frac{2^{23} \cdot \{ [I_{IDAC1} \cdot (R_{LEAD1} + R_{RTD}) - (I_{IDAC2} \cdot R_{LEAD2})] + [I_{IDAC2} \cdot (R_{LEAD1} + R_{RTD}) - (I_{IDAC1} \cdot R_{LEAD2})] \}}{2 \cdot [(I_{IDAC1} + I_{IDAC2}) \cdot R_{REF}]} \quad (10)$$

Then combine $(I_{IDAC1} + I_{IDAC2})$ terms:

$$\text{Averaged output code} = 2^{23} \cdot [(I_{IDAC1} + I_{IDAC2}) \cdot (R_{LEAD1} + R_{RTD}) - (I_{IDAC1} + I_{IDAC2}) \cdot R_{LEAD2}] / [2 \cdot (I_{IDAC1} + I_{IDAC2}) \cdot R_{REF}] \quad (11)$$

Then cancel the $I_{IDAC1} + I_{IDAC2}$ terms and set $R_{LEAD1} = R_{LEAD2} = R_{LEAD}$ to get the following equations:

$$\text{Averaged output code} = 2^{23} \cdot [(R_{LEAD} + R_{RTD}) - R_{LEAD}] / (2 \cdot R_{REF}) \quad (12)$$

After this, the R_{LEAD} terms are cancelled as well.

$$\text{Averaged output code} = 2^{23} \cdot R_{RTD} / (2 \cdot R_{REF}) = 2^{22} \cdot R_{RTD} / R_{REF} \quad (13)$$

Going through the results to Equation 13, it is not important that I_{IDAC1} and I_{IDAC2} are not equal, it is only important that I_{IDAC1} and I_{IDAC2} are the same values after they are swapped. If they are the same, then the $(I_{IDAC1} + I_{IDAC2})$ terms cancel out.

There may still be errors in the system. Here, R_{LEAD1} and R_{LEAD2} are assumed to be the same. If they are different, this becomes an error. Also, if there are leakage currents in the measurement (from TVS or other protection diodes for example), then the leakage contributes to the error.

1.5 Design Considerations

Designing an RTD measurement system requires balancing several different design goals and circuit considerations. After selecting components and excitation magnitude, the designer must verify that the design fits in the operating range of the ADC which includes reference voltage magnitude, the input range of the PGA, and the compliance voltage of any excitation current sources. This section is a basic guide to setting the parameters of operation to design an RTD measurement system with precision ADCs. The basic ratiometric measurement shown in [Figure 1-4](#) will be the starting point for an RTD measurement circuit.

Later sections describe different circuit topologies used for measurement of different RTD wiring configurations. By extension, calculations found here can be applied to different topologies shown later.

1.5.1 Identify the RTD Range of Operation

Start by determining the expected temperature measurement range required for the system, because this will set the range of RTD resistance measurement. As an example, start with a PT100 RTD. The resistance of a PT100 RTD over temperature was shown in [Figure 1-1](#). If the required system temperature measurement range is -200°C to 850°C , this requires the full measurement range of a PT100 RTD. With this temperature range, the RTD would have an equivalent resistance range of $20\ \Omega$ to $400\ \Omega$. Use this resistance range to start the design of the measurement system. Determining the temperature range and then the RTD resistance range helps set the excitation current, gain, and the reference resistance in the design.

1.5.2 Set the Excitation Current Sources and Consider RTD Self Heating

Many precision ADCs used for RTD measurement will have programmable excitation current sources (IDACs) in several magnitudes. A precision ADC device may have a matched pair of IDACs used for excitation. These IDACs can be set to currents of 10, 50, 100, 250, 500, 750, 1000, 1500, and 2000 μA . Excitation currents are used to drive both the RTD, the reference resistance and biasing resistors for some designs.

For the best noise performance, maximize the excitation current used for the RTD and reference resistance excitation. However, most excitation currents should be kept lower than 1 mA because of self heating. Because there is current running through the RTD, the RTD itself will dissipate power through heat. This self heating will cause an error in the measurement. The change in temperature (ΔT) is determined by the power dissipation of the RTD divided by the self-heating coefficient E , in $\text{mW}/^{\circ}\text{C}$. This change in temperature becomes a temperature measurement error and is shown in [Equation 14](#).

$$\Delta T = (I_{\text{IDAC}})^2 \cdot R_{\text{RTD}} / E \quad (14)$$

The typical range of RTD self-heating coefficients is $2.5\ \text{mW}/^{\circ}\text{C}$ for small, thin-film elements and $65\ \text{mW}/^{\circ}\text{C}$ for larger, wire-wound elements. With 1-mA excitation at the maximum RTD resistance value and a larger self-heating coefficient, the power dissipation in the RTD is less than 0.4 mW and will keep the measurement errors due to self-heating to less than 0.01°C . Self-heating coefficients will vary with RTD construction and the measurement medium (in air or in water, for example). Consult the RTD manufacturer data sheet for sensor characteristics.

Referring back to [Figure 1-4](#), this topology uses a single IDAC current source. Other topologies may use matched sources to for lead current calculation.

1.5.3 Set Reference Voltage and PGA Gain

After selecting the IDAC current, use the maximum reference resistor possible, but consider several factors in the setting the reference. The reference voltage must be within the minimum and maximum reference voltages for operation. Many ADCs will have a minimum value of the reference of 0.5 V. Some devices will have a reference maximum of $\text{AVDD} - \text{AVSS}$, while others may have a lower maximum of $\text{AVDD} - \text{AVSS} - 1\ \text{V}$. Consult the ADC data sheet for more specifications on the external reference input range.

A good selection for the reference voltage is using a value close to the midpoint of $\text{AVDD} - \text{AVSS}$. Often, this reference voltage is used to set up the common-mode voltage for the input measurement. PGA amplification may be limited by its input range and output swing. By setting the input common-mode voltage to the midpoint in the supplies, the PGA will have the maximum range possible. Many precision ADCs have a PGA that can amplify small input signals. These PGAs often will have gains from 1 V/V to 128 V/V in factors of 2.